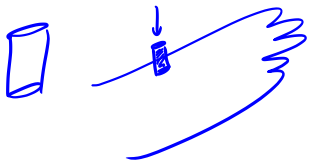
DMF DENTAL TRAINING CENTER: FINAL COACHING

General Histology and Microbiology

By: Mickhail Q. Francisco, DMD

1. Small, spherical-shaped membrane-bound organelles formed from the endoplasmic reticulum which contain oxidative enzymes. They detoxify specific harmful substances either produced by the cell or taken into the cell.
   a. Lysosomes – vesicles that contain enzymes
   b. Peroxisomes
   c. Ribosomes – protein synthesis
   d. Vesicles – contain products of cells
2. A 17-year-old patient presents with a persistent white lesion on the lateral border of the tongue that has failed to resolve after several weeks of observation. The oral surgeon decides to remove the entire lesion and submit it for histopathologic examination. Which type of biopsy procedure is being performed in this case?
   a. Incisional biopsy = portion of lesion (large lesion >1cm)
   b. Excisional biopsy = entire lesion (small lesion <1cm)
   c. Punch biopsy
   d. Aspiration biopsy
3. A dermatologist evaluates a patient with a small mucosal lesion and uses a circular blade instrument to obtain a cylindrical specimen that includes the epithelium and underlying connective tissue for microscopic examination. Which biopsy technique is being described?
   a. Needle aspiration biopsy (FNAB) → Fluid
   b. Excisional biopsy
   c. Punch biopsy
   d. Incisional biopsy



4. It is a process in which the cell uses lysosomes to dispose excess or nonfunctioning organelles or membranes
   a. Autophagy – normal cellular process
   b. Apoptosis – normal cellular process = programmed cell death (mitochondria)
   c. Pyroptosis – pathologic (inflammatory cell death – infection)
   d. Autolysis – lysosomes leak → cell destruction (due to cell injury – HOT/Cold/Trauma/Chemical)
5. Which of the following is ~~not an~~ incorrect statement regarding eukaryotes and prokaryotes?
   a. Eukaryotes and prokaryotes contain mitochondria (F)
   b. Prokaryotes and eukaryotes have nucleus (F)
   c. Eukaryotes have diploid chromosome while prokaryotes have haploid

2 sets 1 set

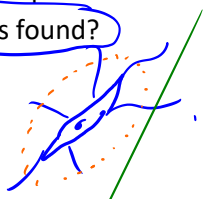
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d. Organelles of prokaryotes are membrane bound

6. A dental student is reviewing the microbiology of aggressive periodontitis in adolescents and is asked to identify the microorganism most strongly associated with localized juvenile periodontitis. Which of the following is the correct answer?
   a. Porphyromonas gingivalis
   b. Aggregatibacter actinomycetemcomitans
   c. Fusobacterium nucleatum
   d. Prevotella intermedia

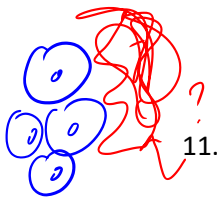
7. Umbrella cells are found lining which of the following tissues?
   a. Respiratory tract
   b. Epidermis
   c. Urinary tract
   d. Esophagus

8. During a histology examination, a student is asked to identify the location of mature bone cells that maintain the bone matrix and occupy small spaces within calcified bone tissue. These cells are known as osteocytes. Where are osteocytes found?
   a. Endosteum
   b. Periosteum
   c. Lacunae
   d. Howship's Lacunae



9. Glands of the oral cavity all have major basic components. One component that is not a constant feature of all glands is the ____
   a. Stroma
   b. Capsule
   c. Vascular element
   d. Parenchymal cells

10. All are characteristics of fungus except one. Which is it?
    a. Dimorphic
    b. Prokaryotic
    c. Septated
    d. Sporulate



11. The extracellular matrix of the connective tissue is typically made up of which of the following?
    a. ~~Resident cells~~ and Water
    b. Plasma and ~~Resident cells~~
    c. Water and Plasma
    d. Protein fibers and Ground substance

12. In the nitrogen cycle, microorganisms decompose proteins and other nitrogen-containing organic compounds, resulting in the release of ammonia into the environment. Which biological process is responsible for this conversion?
    a. Nitrification

b. Ammonification
c. Denitrification
d. Nitrogen fixation

13. This is the main class of serum antibodies found in birds, amphibians and reptiles
    a. IgG
    b. IgY
    c. IgA
    d. IgM
14. This is the main class of antibodies found in egg yolk
    a. IgG
    b. IgY
    c. IgA
    d. IgM
15. It is a prokaryotic organelle that functions similarly to eukaryotic mitochondria
    a. Lysosome
    b. Oxysome
    c. Ribosome
    d. Mesosome
16. A patient presents with oral thrush characterized by white plaques on the oral mucosa. Laboratory examination reveals budding yeast cells and pseudohyphae. Which microorganism is most likely responsible for this infection?
    a. Clostridium botulinum
    b. Candida albicans
    c. Mycobacterium tuberculosis
    d. Vibrio cholerae
17. A hospital infection control committee is discussing methods for sterilizing heat-sensitive medical instruments that cannot tolerate steam sterilization. Which chemical sterilizing gas has traditionally been one of the most widely used agents for this purpose?
    a. Chlorine dioxide gas
    b. Formaldehyde vapor
    c. Hydrogen peroxide gas plasma
    d. Ethylene oxide gas
18. A child diagnosed with acute herpetic gingivostomatitis should be treated with which of the following drugs?
    a. Oral antivirals
    b. Antibiotics
    c. Observe
    d. Prescribe mouth wash or oral rinse

19. A patient develops a painless ulcer and is subsequently diagnosed with syphilis. The physician explains that the disease is caused by a highly motile spirochete bacterium. Which organism is the etiologic agent of syphilis?
    a. Treponema denticola – periodontitis (NUG)
    b. Treponema pallidum
    c. Borrelia burgdorferi – lyme do. (Tick bites) → Bull's eye lesion (> scary)
    d. Leptospira interrogans – leptospirosis

20. During a microbiology lecture, students learn that certain organisms are incapable of synthesizing their own food and must obtain nutrients and organic compounds from other living organisms or environmental sources. What term best describes these organisms?
    a. Heterotrophs — Hetero – other/another/different
    b. Autotrophs = (SCP) = can create their own food
    c. Phototrophs — light
    d. Lithotrophs — minerals
    Chemotrophs – chemical (minerals)

21. It is the main component of dental plaque
    a. Microorganism
    b. Dextrans – sticky sugar (adhesive)
    c. Levans
    d. Proteins

22. A professor asks the class to identify the most essential structure in a prokaryotic cell that contains the genetic material responsible for controlling cellular activities, replication, and inheritance. Which structure should be selected?
    a. Nucleoid
    b. Cell wall
    c. Cell membrane
    d. Capsule

23. It is responsible for the solidification of the culture media.
    a. Electrolytes – Na Cl K Mg Ca
    b. Agar – seaweed
    c. Water – Hydrogen + Oxygen
    d. Peptone – amino acids

24. All of the following are types of biopsies, except one. Which is it?
    a. Aspiration
    b. Excision
    c. Immunofluorescent – "Identification of cells/microbes/parasites"
    d. Punch

25. During a microbiology laboratory exercise, a student is instructed to prepare a culture medium containing tryptone, glucose, and yeast extract (TGYE) to support the growth of a wide variety of nonfastidious microorganisms. How is this culture medium best classified?
    ↓
    x picky
    x specific nutritional requirements



a. Selective medium only
b. Differential medium only
c. General-purpose enriched medium
d. Transport medium

26. While examining a bacterial smear under a compound light microscope, a student first looks through the lens located nearest the eye, which further magnifies the image produced by the objective lens. Which component of the light microscope serves as the primary magnifying lens?
    a. Objective lens
    b. Eyepiece
    c. Condenser lens
    d. ~~Magnifying lens~~

27. A microbiologist observes spherical bacterial cells arranged in irregular grape-like clusters under the microscope. This arrangement is characteristic of which bacterial genus morphology?
    a. Staphylococci
    b. Streptococci
    c. Bacilli
    d. Spirochetes

28. Hydrostatic pressure of the blood is caused by the pumping action of the heart, which forces water out across the capillary wall; While the colloid osmotic pressure of blood produced by plasma proteins such as albumin draws water back into the interstitial tissues.
    a. Both statements are true
    b. Both statements are false
    c. The first statement is true; the second is false
    d. The first statement is false; the second is true

29. When identifying bacteria using a microscope, which criteria is not used?
    a. Cellular arrangement
    b. Pigmentation
    c. Shape
    d. Texture

30. A patient suspected of having tuberculosis undergoes laboratory testing. The causative organism possesses a waxy cell wall rich in mycolic acids and is not reliably demonstrated using a simple Gram stain. Which staining method is most appropriate for visualizing Mycobacterium species?
    a. Gram stain
    b. Giemsa stain
    c. Acid-fast stain
    d. India ink stain

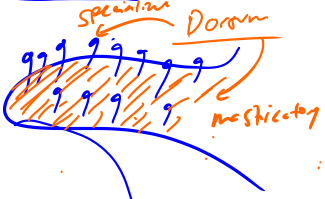
31. Brown adipose cells are specialized for long-term energy storage. White adipose cells are primarily for thermogenesis.
    a. Both statements are true

b. Both statements are false
c. The first statement is true; the second is false
d. The first statement is false; the second is true

32. A histology instructor asks students to identify the outermost layer of the skin that provides a protective barrier against physical, chemical, and microbial injury. Which structure should be selected?
a. Epidermis
b. Dermis
c. Hypodermis
d. Subcutaneous tissue

33. Among the following, which is the not an essential component of a simple culture media?
a. Water
b. Blood
c. Amino acid
d. Peptone

34. During an oral histology practical examination, a student is asked to identify the type of oral mucosa that covers the dorsal surface of the tongue and contains taste buds associated with the lingual papillae. Which type of mucosa is this?
a. Lining mucosa
b. Masticatory mucosa
c. Specialized mucosa
d. Keratinized mucosa



35. A clinical microbiologist is attempting to differentiate two Gram-positive cocci isolated from a patient's oral infection. Which laboratory finding would most definitively distinguish Staphylococcus species from Streptococcus species?
a. Streptococcus is catalase-positive and grows in clusters.
b. Staphylococcus is catalase-positive and typically grows in clusters.
c. Streptococcus is coagulase-positive and grows in clusters.
d. Staphylococcus is oxidase-positive and grows in pairs.

36. Hyphae are filaments made up of fungi cells. The 3 types of Hyphae are Septate, Aseptate and Pseudohyphae
a. The first statement is true and the second statement is false.
b. The first statement is false and the second statement is true.
c. Both statements are true
d. Both statements are false

37. This is an infectious disease-causing protein that is believed to be the smallest infectious particle.
a. Bacteria
b. Virus
c. Fungi
d. Prion

38. A dentist explains to a patient that one of the hard tissues of the tooth lacks living cells and therefore cannot regenerate once it is destroyed by caries, trauma, or wear. Which dental tissue is being described?
    a. Enamel
    b. Cementum
    c. Dentin
    d. Alveolar bone

39. A patient presents with an acute bacterial infection, and laboratory testing reveals a marked increase in one of the major acute-phase proteins synthesized by the liver in response to inflammatory cytokines such as interleukin-6. Which protein is most commonly elevated during inflammation and is widely used as a clinical marker of inflammatory activity?
    a. C-reactive protein
    b. Albumin – Blood
    c. Transferrin – Iron carrier
    d. Hemoglobin – Oxygen carrier / CO_2 / other gas

40. A pathologist examines a tissue specimen from a patient with a long-standing inflammatory lesion. The microscopic examination reveals the predominant presence of cells responsible for phagocytosis, antigen presentation, and immune regulation, along with cells involved in adaptive immunity. Which cells are most characteristic of chronic inflammation?
    a. Neutrophils and platelets
    b. Macrophages and lymphocytes, Plasma cells
    c. Eosinophils and basophils
    d. Mast cells and erythrocytes

41. Slime layers and capsules are types of _____
    a. Bacteria
    b. Glycocalyx – extracellular material secreted by bacteria
    c. Cell wall
    d. Cell membrane

42. During a histology practical examination, a student is asked to identify the nonciliated, dome-shaped epithelial cells found in the terminal bronchioles that secrete protective substances and help detoxify harmful inhaled materials. What are these cells called?
    a. Goblet cells
    b. Basal cells
    c. Club cells / Clara cells
    d. Type II pneumocytes

43. Endotoxin is released when the cell wall of _____ are disrupted.
    a. Gram-positive bacteria
    b. Gram-negative bacteria
    c. Acid-fast bacteria

d. All of the above

44. Gram-staining procedure involves all of the following solutions, except one. Which is it?
    a. Crystal violet
    b. Methylene blue
    c. Iodine
    d. Alcohol

45. It is an inflammatory cell death that is not part of normal cellular process but rather due to microbial infection. This process is important for the inflammatory response of the body.
    a. Autophagy
    b. Apoptosis
    c. Pyroptosis
    d. Autolysis

46. During an anatomy lecture, students are asked to identify the structural unit that forms compact bone. These units consist of concentric layers of mineralized matrix arranged around a central canal. What are these concentric layers called?
    a. Trabeculae
    b. Osteons
    c. Lamellae
    d. Canaliculi

47. Meningitis in infants and children is commonly due to which of the following organisms?
    a. Neisseria meningitidis
    b. Streptococcus pneumoniae
    c. Escherichia coli
    d. Haemophilus influenzae

48. A hematology professor asks the class where granulocytes such as neutrophils, eosinophils, and basophils are primarily produced before entering the bloodstream. Which of the following is the correct site of formation?
    a. Spleen
    b. Lymph nodes
    c. Thymus
    d. Red bone marrow

49. Among the following microorganisms, which cannot generate their own ATP and depends their energy source solely to their host cells
    a. Rickettsia
    b. Streptococcus
    c. Pseudomonas
    d. Porphyromonas

50. A young child presents with fever, sore throat, and multiple vesicular lesions on the soft palate and posterior oropharynx. The physician diagnoses herpangina, a viral disease commonly seen

during the summer months. Which microorganism is most commonly responsible for this condition?

a. Epstein-Barr virus
b. Herpes simplex virus type 1
c. Coxsackie A virus
d. Varicella-zoster virus

51. During the history of antimicrobial drug discovery, Alexander Fleming observed that a mold contaminating a bacterial culture inhibited bacterial growth, leading to the discovery of penicillin. From which organism is penicillin originally derived?

a. Aspergillus fumigatus
b. Candida albicans
c. Penicillium notatum
d. Rhizopus stolonifer

52. When microorganisms are placed in an environment that provides optimum growth requirements

a. The generation time is slower
b. The generation is faster
c. The generation time remains the same
d. The endospores are formed

53. This is the enzyme used by HIV to convert itself from RNA virus to DNA form.

a. Reverse transcriptase
b. Transcriptase
c. Polymerase
d. DNA Gyrase

54. A patient is diagnosed with streptococcal pharyngitis caused by Group A β-hemolytic Streptococcus. Which antibiotic is considered the treatment of choice?

a. Metronidazole
b. Ciprofloxacin
c. Penicillin
d. Tetracycline

55. Soap has antiseptic properties primarily because it can_____

a. Kill microorganisms
b. Inhibit the growth of microorganisms
c. Dehydrate microorganisms
d. Remove microorganisms mechanically

56. Which cellular structure is present in all cells and serves as the selectively permeable barrier regulating the movement of substances into and out of the cell?

a. Cell wall
b. Cell membrane
c. Capsule

d. Flagellum

57. Cytopathic effects are commonly associated with which of the following microorganisms?
    a. Mycobacterium
    b. Spirochetes
    c. Prions
    d. Viruses

58. While viewing a stained bacterial specimen under a compound light microscope, a student adjusts a structure located beneath the stage that concentrates and focuses light onto the specimen to improve image clarity and resolution. Which microscope component performs this function?
    a. Aperture
    b. Illuminator
    c. Condenser
    d. Mechanical stage

59. It is not a medium for culturing viruses
    a. Chick embryo
    b. Blood agar
    c. Guinea pigs
    d. Cell culture

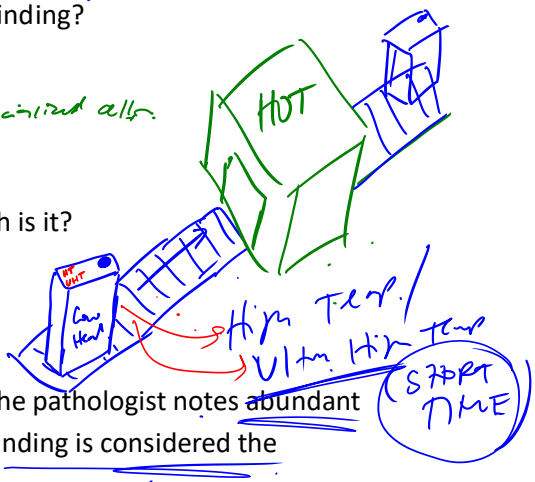
60. A histology professor asks a student to identify the type of epithelium lining most of the conducting portion of the respiratory tract, including the trachea and primary bronchi. Which epithelial type is characteristic of this region?
    a. Stratified squamous keratinized epithelium
    b. Simple cuboidal epithelium
    c. Pseudostratified ciliated columnar epithelium
    d. Transitional epithelium

61. A patient develops a localized abscess following a bacterial infection. Histologic examination demonstrates tissue destruction with the formation of a thick, creamy exudate composed primarily of dead neutrophils and dissolved tissue. What type of necrosis is most characteristic of an abscess?
    a. Caseous necrosis
    b. Fat necrosis
    c. Liquefactive necrosis
    d. Coagulative necrosis

62. Which of the following is a "cell wall-deficient" bacteria?
    a. Rickettsiae
    b. Mycoplasma
    c. Chlamydiae
    d. Ehrlichiae

63. A bodybuilder experiences enlargement of skeletal muscle fibers after months of intensive weight training. Microscopic examination reveals an increase in cell size without an increase in cell number. Which adaptive cellular response best explains this finding?
    a. Hypertrophy
    b. Hyperplasia
    c. Metaplasia
    d. Atrophy

64. The following are advantages of pasteurization, except one. Which is it?
    a. It kills bacteria and spores
    b. It kills all bacteria except thermoduric bacteria
    c. It kills 95% of microorganisms
    d. Tubercle bacilli are destroyed



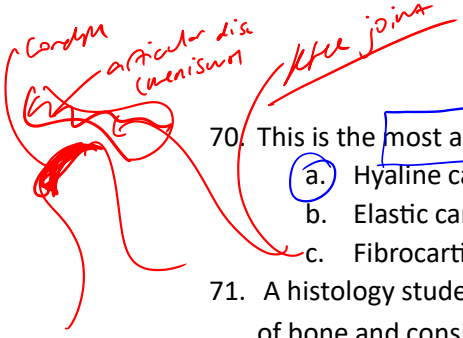
65. During microscopic evaluation of a chronic inflammatory lesion, the pathologist notes abundant collagen deposition replacing normal tissue architecture. Which finding is considered the strongest evidence of tissue repair in chronic inflammation?
    a. Neutrophilic infiltration
    b. Edema formation
    c. Vascular congestion
    d. Fibrosis

66. Mesophilic organisms are microbes that produce optimal growth at what temperatures?
    a. -15°C to 10°C
    b. 0°C to 25°C
    c. 25°C to 40°C
    d. 55°C to 80°C

67. Which statement most accurately describes alveolar bone composition?
    a. It is composed of both compact bone and spongy bone.
    b. It is composed entirely of compact bone.
    c. It is composed entirely of spongy bone.
    d. It contains only woven bone throughout life.

68. The following are examples of dense connective tissues that contains collagen, except one. Which is it?
    a. Tendons
    b. Ligaments
    c. Adipose
    d. Dermis

69. Portal of entry of poliovirus is mainly by which of the following tissues?
    a. GI tract
    b. Upper respiratory tract
    c. Lungs
    d. Skin



70. This is the most abundant type of cartilage which is primarily made up of type II collagen
    a. Hyaline cartilage
    b. Elastic cartilage
    c. Fibrocartilage
71. A histology student is asked to identify the connective tissue layer that covers the outer surface of bone and consists of dense fibrous connective tissue rich in collagen fibers, blood vessels, and osteogenic cells. Which structure is this?
    a. Endosteum
    b. Cementum
    c. Sharpey's fibers
    d. Periosteum
72. Which of the following is a correct description of Virion?
    a. Infectious form of a virus outside the host cell
    b. Smallest type kind of virus
    c. Small particle that mimics a virus
    d. None of the above
73. Refuse storage containers used for the disposal of infectious and pathological waste should be what color?
    a. Black
    b. Green
    c. Blue
    d. Yellow
74. Being infected by a certain type of virus prevents an individual to be infected with another type of virus. This is known as?
    a. Viral immunity
    b. Viral protection
    c. Viral resistance
    d. Viral interference
75. The anterior part of the hard palate near the incisive papilla and rugae contains adipocytes in its submucosa, while the posterior part near the posterolateral corner contains minor salivary glands in its submucosa. The midline raphe region does not contain any submucosa and the mucosa is directly attached to the bone tissue.
    a. Both statements are true
    b. Both statements are false
    c. The first statement is true; the second is false
    d. The first statement is false; the second is true